


TStates : 179 | MCycles : 53 | Time of Execution : 179 µs
Flags : S Z AC P CY

General Regs.:

| A | B | C | D | E | H | L | M | SP | PC | X |
|---|---|---|---|---|---|---|---|---|---|---|
| 00 | 5A | 27 | 00 | 00 | 00 | 00 | 00 | 47FF | 4100 | |

```
LXI SP 47FFh
LDA 4500h
MOV B A
LDA 4501h
MOV C A
MOV A B
ADD C
STA 4502h
MVI A 00h
ACI 00h
STA 4503h
LDA 4500h
MOV B A
LDA 4501h
MOV C A
MOV A B
SUB C
STA 4504h
MVI A 00h
ACI 00h
STA 4505h
HLT
```

Starting Address : 4100

Comments :

**User Data Grid**

| Address | Data |
|---|---|
| 4500 | 5A |
| 4501 | 27 |
| 4502 | 81 |
| 4503 | 00 |
| 4504 | 33 |
| 4505 | 00 |
| 4510 | 34 |
| 4511 | 12 |
| 4512 | 21 |
| 4513 | 43 |
| 4516 | 00 |
| 4517 | 00 |
| 4518 | 00 |
| 4519 | 00 |
| 451A | 00 |

**Hex Code Grid**

| Address | Data |
|---|---|
| 4100 | 31 |
| 4101 | FF |
| 4102 | 47 |
| 4103 | 3A |
| 4104 | 00 |
| 4105 | 45 |
| 4106 | 47 |
| 4107 | 3A |
| 4108 | 01 |
| 4109 | 45 |
| 410A | 4F |
| 410B | 78 |
| 410C | 81 |
| 410D | 32 |
| 410E | 02 |
| 410F | 45 |
| 4110 | 3E |
| 4111 | 00 |

1.Data Transfer Group :

| | | | |
|---|---|---|---|
| MOV | MVI | LDA | STA |
| LHLD | SHLD | LXI | XCHG |
| LDAX | STAX | IN | OUT |

2.Arithmetic Group :

| | | | | |
|---|---|---|---|---|
| ADD | ADI | ADC | ACI | DAA |
| SUB | SUI | SBB | SBI | DAD |
| INR | DCR | INX | DCX | |

3.Logical Group :

| | | | | |
|---|---|---|---|---|
| ANA | ANI | XRA | XRI | CMA |
| ORA | ORI | CMP | CPI | CMC |
| RLC | RRC | RAL | RAR | STC |

4.Branch and Machine Control Group

| | | | | |
|---|---|---|---|---|
| JMP | J_ | CALL | C_ | |
| RET | R_ | EI | DI | PUSH |
| PCHL | RST | RIM | SIM | POP |
| XTHL | SPHL | NOP | HLT | |

Auto Insert | Insert | Cancel

Instruction Information :

Code Pad




TStates : 231 | MCycles : 69 | Time of Execution : 231 µs

Flags : S Z AC P CY

General Regs.: A 01 | B C 12 43 | D E 12 34 | H L 55 55 | M 00 | SP 0000 | PC 4100

```
LHLD 4510h
XCHG
LHLD 4512h
DAD D
SHLD 4514h
MVI A 00h
ACI 00h
STA 4516h
LDA 4510h
MOV B A
LDA 4512h
MOV C A
MOV A B
SUB C
STA 4518h
LDA 4511h
MOV B A
LDA 4513h
MOV C A
MOV A B
SBB C
STA 4519h
MVI A 00h
ACI 00h
STA 451Ah
HLT
```

Starting Address : 4100

Comments :

**User Data Grid**

| Address | Data |
|---|---|
| 4510 | 34 |
| 4511 | 12 |
| 4512 | 21 |
| 4513 | 43 |
| 4514 | 55 |
| 4515 | 55 |
| 4516 | 00 |
| 4518 | 13 |
| 4519 | CF |
| 451A | 01 |

**Hex Code Grid**

| Address | Data |
|---|---|
| 4100 | 2A |
| 4101 | 10 |
| 4102 | 45 |
| 4103 | EB |
| 4104 | 2A |
| 4105 | 12 |
| 4106 | 45 |
| 4107 | 19 |
| 4108 | 22 |
| 4109 | 14 |
| 410A | 45 |
| 410B | 3E |
| 410C | 00 |
| 410D | CE |
| 410E | 00 |
| 410F | 32 |
| 4110 | 16 |
| 4111 | 45 |

1.Data Transfer Group :

| | | | |
|---|---|---|---|
| MOV | MVI | LDA | STA |
| LHLD | SHLD | LXI | XCHG |
| LDAX | STAX | IN | OUT |

2.Arithmatic Group :

| | | | | |
|---|---|---|---|---|
| ADD | ADI | ADC | ACI | DAA |
| SUB | SUI | SBB | SBI | DAD |
| INR | DCR | INX | DCX | |

3.Logical Group :

| | | | | |
|---|---|---|---|---|
| ANA | ANI | XRA | XRI | CMA |
| ORA | ORI | CMP | CPI | CMC |
| RLC | RRC | RAL | RAR | STC |

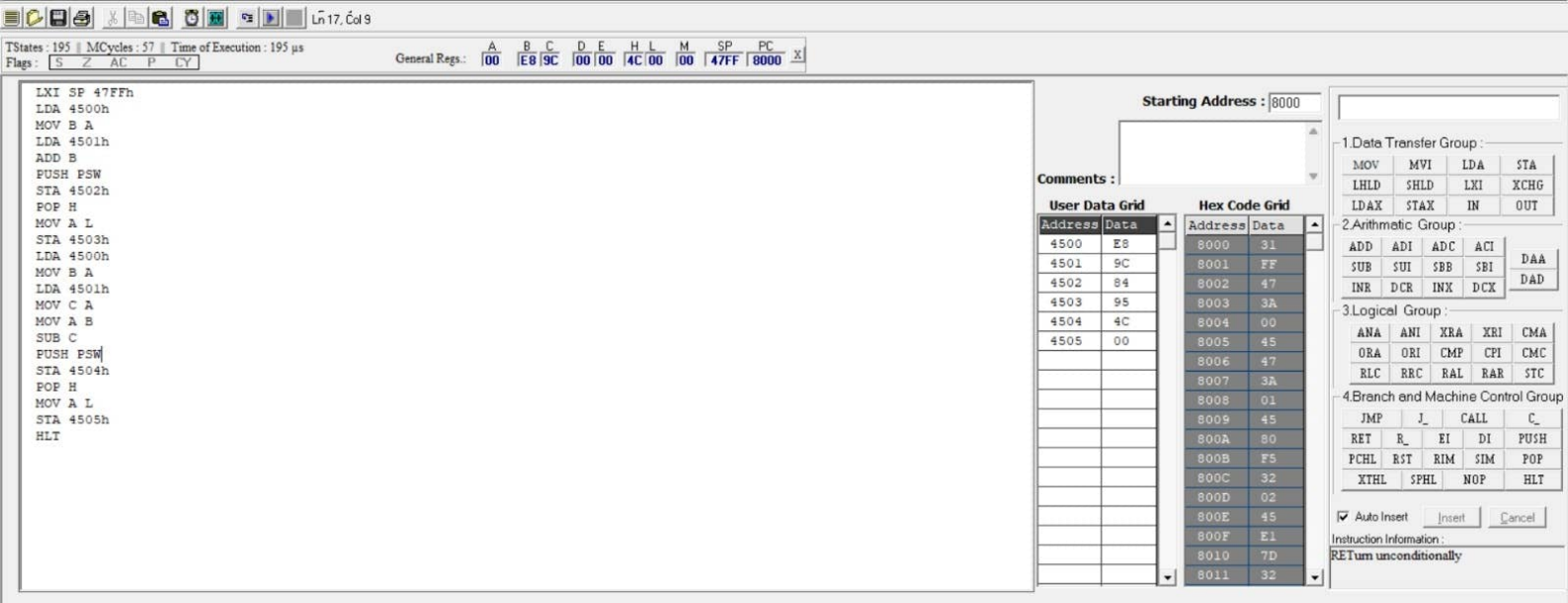
4.Branch and Machine Control Group

| | | | | |
|---|---|---|---|---|
| JMP | J_ | CALL | C_ | |
| RET | R_ | EI | DI | PUSH |
| PCHL | RST | RIM | SIM | POP |
| XTHL | SPHL | NOP | HLT | |

Auto Insert | Insert | Cancel

Instruction Information :

MOVe -- copy from source to destination


Ln 17, Col 9
TStates : 195 | MCycles : 57 | Time of Execution : 195 µs
Flags : S Z AC P CY
General Regs.: A 00 B C E8 9C D E 00 00 H L 4C 00 M 00 SP 47FF PC 8000 X
LXI SP 47FFh
LDA 4500h
MOV B A
LDA 4501h
ADD B
PUSH PSW
STA 4502h
POP H
MOV A L
STA 4503h
LDA 4500h
MOV B A
LDA 4501h
MOV C A
MOV A B
SUB C
PUSH PSW
STA 4504h
POP H
MOV A L
STA 4505h
HLT
Starting Address : 8000
Comments :
User Data Grid
Address Data
4500 E8
4501 9C
4502 84
4503 95
4504 4C
4505 00
Hex Code Grid
Address Data
8000 31
8001 FF
8002 47
8003 3A
8004 00
8005 45
8006 47
8007 3A
8008 01
8009 45
800A 80
800B F5
800C 32
800D 02
800E 45
800F E1
8010 7D
8011 32
1.Data Transfer Group :
MOV MVI LDA STA
LHLD SHLD LXI XCHG
LDAX STAX IN OUT
2.Arithmatic Group :
ADD ADI ADC ACI
SUB SUI SBB SBI
INR DCR INX DCX
DAA
DAD
3.Logical Group :
ANA ANI XRA XRI CMA
ORA ORI CMP CPI CMC
RLC RRC RAL RAR STC
4.Branch and Machine Control Group
JMP J_ CALL C_
RET R_ EI DI PUSH
PCHL RST RIM SIM POP
XTHL SPHL NOP HLT
Auto Insert
Insert
Cancel
Instruction Information :
RETurn unconditionally




TStates : 411 | MCycles : 119 | Time of Execution : 411 µs

Flags : S Z AC P CY

General Regs.: A 03 | B C 06 03 | D E 00 14 | H L 00 78 | M 00 | SP 0000 | PC 8000

```
LDA 4520h
MOV E A
MVI D 00h
LDA 4521h
MOV C A
LXI H 0000h
MOV A C
ORA A
JZ MULT_DONE
MULT_LOOP: DAD D
DCR C
JNZ MULT_LOOP
MULT_DONE: SHLD 4522h
LDA 4525h
MOV B A
MOV A B
ORA A
JZ DIV_ZERO
LDA 4524h
MVI C 00h
DIV_LOOP: CMP B
JC DIV_DONE
SUB B
INR C
JMP DIV_LOOP
DIV_DONE: STA 4527h
MOV A C
STA 4526h
HLT
DIV_ZERO: MVI A 0Fh
STA 4540h
HLT
```

Starting Address : 8000

Comments :

**User Data Grid**

| Address | Data |
|---|---|
| 4520 | 14 |
| 4521 | 06 |
| 4524 | 14 |
| 4525 | 06 |
| 4522 | 78 |
| 4523 | 00 |
| 4526 | 03 |
| 4527 | 02 |

**Hex Code Grid**

| Address | Data |
|---|---|
| 8000 | 3A |
| 8001 | 20 |
| 8002 | 45 |
| 8003 | 5F |
| 8004 | 16 |
| 8005 | 00 |
| 8006 | 3A |
| 8007 | 21 |
| 8008 | 45 |
| 8009 | 4F |
| 800A | 21 |
| 800B | 00 |
| 800C | 00 |
| 800D | 79 |
| 800E | B7 |
| 800F | CA |
| 8010 | 17 |
| 8011 | 80 |

1.Data Transfer Group :

| | | | |
|---|---|---|---|
| MOV | MVI | LDA | STA |
| LHLD | SHLD | LXI | XCHG |
| LDAX | STAX | IN | OUT |

2.Arithmetic Group :

| | | | | |
|---|---|---|---|---|
| ADD | ADI | ADC | ACI | DAA |
| SUB | SUI | SBB | SBI | DAD |
| INR | DCR | INX | DCX | |

3.Logical Group :

| | | | | |
|---|---|---|---|---|
| ANA | ANI | XRA | XRI | CMA |
| ORA | ORI | CMP | CPI | CMC |
| RLC | RRC | RAL | RAR | STC |

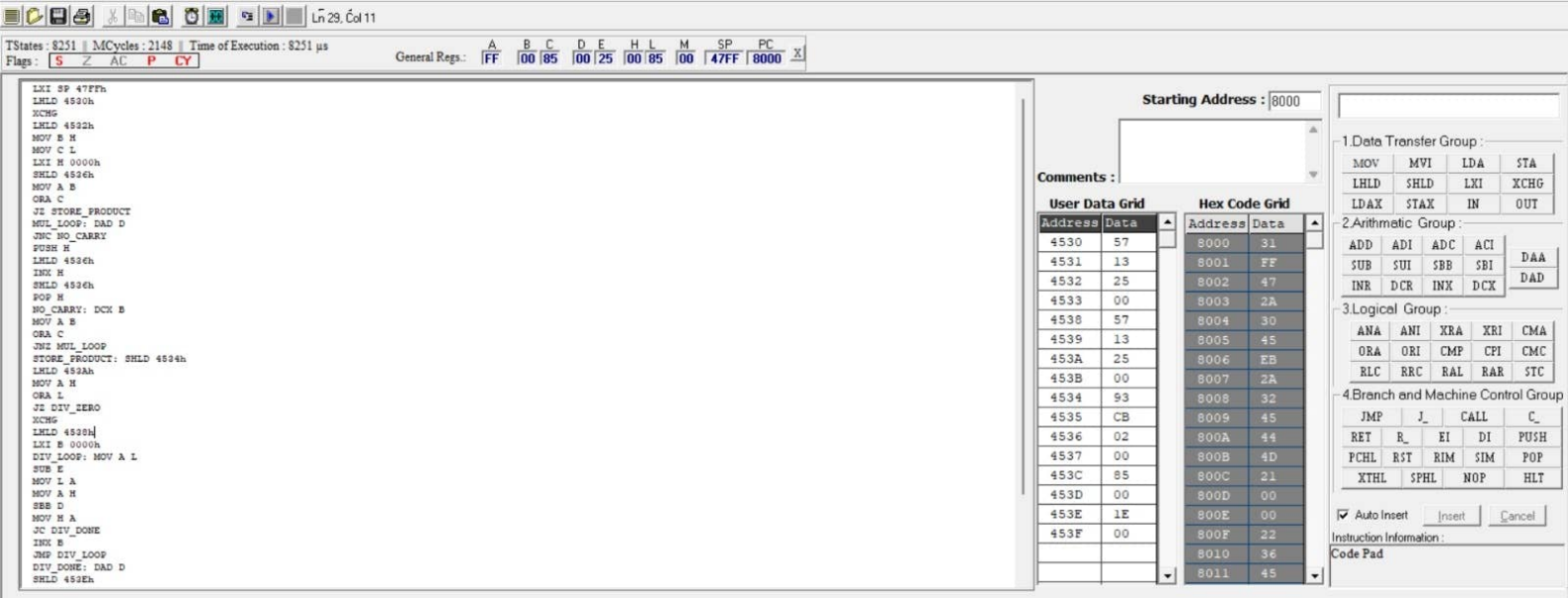
4.Branch and Machine Control Group

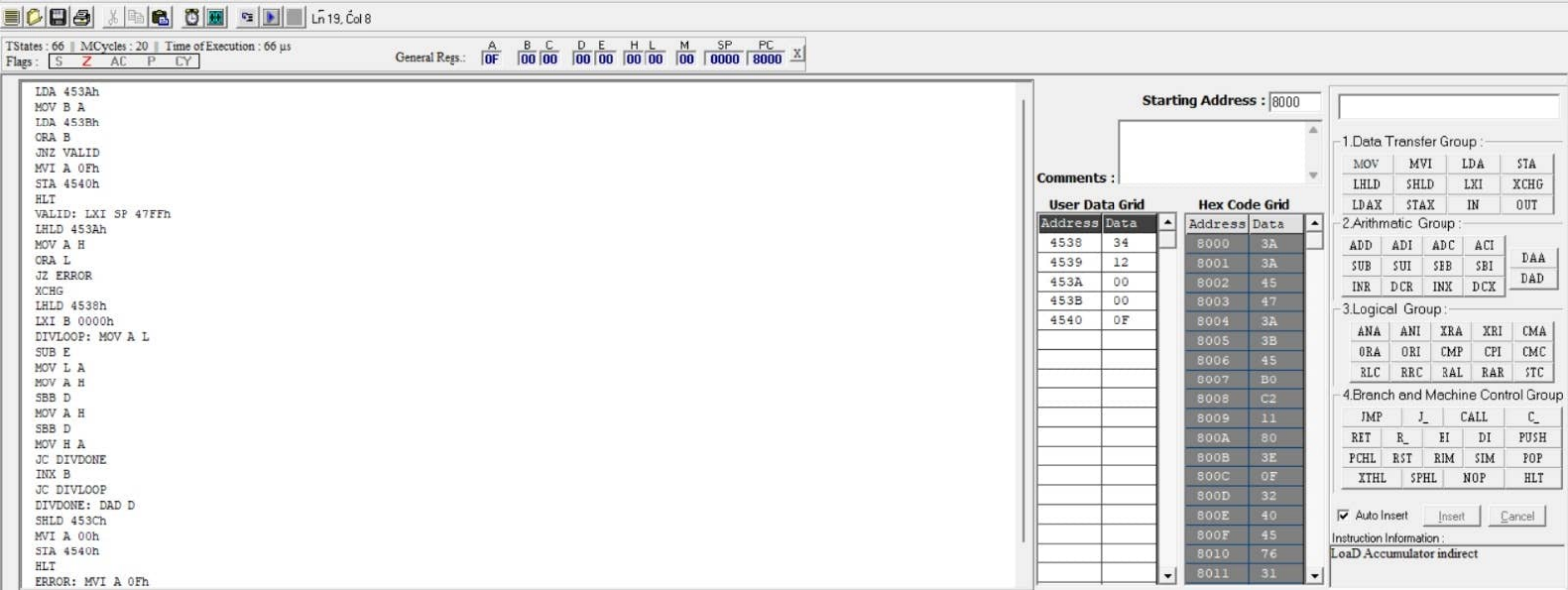
| | | | | |
|---|---|---|---|---|
| JMP | J_ | CALL | C_ | |
| RET | R_ | EI | DI | PUSH |
| PCHL | RST | RIM | SIM | POP |
| XTHL | SPHL | NOP | HLT | |

Auto Insert | Insert | Cancel

Instruction Information :

LoaD H and L register direct


Ln 29, Col 11
TStates : 8251 | MCycles : 2148 | Time of Execution : 8251 µs
Flags : S Z AC P CY
General Regs.: A FF B C 00 85 D E 00 25 H L 00 85 M 00 SP 47FF PC 8000
LXI SP 47FFh
LHLD 4530h
XCHG
LHLD 4532h
MOV B M
MOV C L
LXI H 0000h
SHLD 4536h
MOV A B
ORA C
JZ STORE_PRODUCT
MUL_LOOP: DAD D
JNC NO_CARRY
PUSH H
LHLD 4536h
INX H
SHLD 4536h
POP H
NO_CARRY: DCX B
MOV A B
ORA C
JNZ MUL_LOOP
STORE_PRODUCT: SHLD 4534h
LHLD 453Ah
MOV A H
ORA L
JZ DIV_ZERO
XCHG
LHLD 4538h
LXI B 0000h
DIV_LOOP: MOV A L
SUB E
MOV L A
MOV A H
SBB D
MOV H A
JC DIV_DONE
INX B
JMP DIV_LOOP
DIV_DONE: DAD D
SHLD 453Eh
Starting Address : 8000
Comments :
User Data Grid
Address Data
4530 57
4531 13
4532 25
4533 00
4538 57
4539 13
453A 25
453B 00
4534 93
4535 CB
4536 02
4537 00
453C 85
453D 00
453E 1E
453F 00
Hex Code Grid
Address Data
8000 31
8001 FF
8002 47
8003 2A
8004 30
8005 45
8006 EB
8007 2A
8008 32
8009 45
800A 44
800B 4D
800C 21
800D 00
800E 00
800F 22
8010 36
8011 45
1.Data Transfer Group :
MOV MVI LDA STA
LHLD SHLD LXI XCHG
LDAX STAX IN OUT
2.Arithmetic Group :
ADD ADI ADC ACI
SUB SUI SBB SBI
INR DCR INX DCX
DAA
DAD
3.Logical Group :
ANA ANI XRA XRI CMA
ORA ORI CMP CPI CMC
RLC RRC RAL RAR STC
4.Branch and Machine Control Group
JMP J_ CALL C_
RET R_ EI DI PUSH
PCHL RST RIM SIM POP
XTHL SPHL NOP HLT
Auto Insert
Insert
Cancel
Instruction Information :
Code Pad


Ln 19, Col 8
TStates : 66 | MCycles : 20 | Time of Execution : 66 µs
Flags : S Z AC P CY
General Regs.: A 0F | B C 00 00 | D E 00 00 | H L 00 00 | M 00 | SP 0000 | PC 8000
LDA 453Ah
MOV B A
LDA 453Bh
ORA B
JNZ VALID
MVI A 0Fh
STA 4540h
HLT
VALID: LXI SP 47FFh
LHLD 453Ah
MOV A H
ORA L
JZ ERROR
XCHG
LHLD 4538h
LXI B 0000h
DIVLOOP: MOV A L
SUB E
MOV L A
MOV A H
SBB D
MOV A H
SBB D
MOV H A
JC DIVDONE
INX B
JC DIVLOOP
DIVDONE: DAD D
SHLD 453Ch
MVI A 00h
STA 4540h
HLT
ERROR: MVI A 0Fh
Starting Address : 8000
Comments :
User Data Grid
Address Data
4538 34
4539 12
453A 00
453B 00
4540 0F
Hex Code Grid
Address Data
8000 3A
8001 3A
8002 45
8003 47
8004 3A
8005 3B
8006 45
8007 B0
8008 C2
8009 11
800A 80
800B 3E
800C 0F
800D 32
800E 40
800F 45
8010 76
8011 31
1.Data Transfer Group :
MOV MVI LDA STA
LHLD SHLD LXI XCHG
LDAX STAX IN OUT
2.Arithmatic Group :
ADD ADI ADC ACI
SUB SUI SBB SBI
INR DCR INX DCX
DAA
DAD
3.Logical Group :
ANA ANI XRA XRI CMA
ORA ORI CMP CPI CMC
RLC RRC RAL RAR STC
4.Branch and Machine Control Group
JMP J_ CALL C_
RET R_ EI DI PUSH
PCHL RST RIM SIM POP
XTHL SPHL NOP HLT
Auto Insert
Insert
Cancel
Instruction Information :
LoaD Accumulator indirect




TStates : 252 | MCycles : 74 | Time of Execution : 252 µs
Flags : S Z AC P CY

General Regs.: A 07 | B C 00 00 | D E 00 FB | H L 06 DD | M 00 | SP 0000 | PC 4300 | X

```
LDA 4520h
MOV E A
MVI D 00h
LDA 4521h
MOV C A
LXI H 0000h
MOV A C
ORA A
JZ MULTDONE
MULTLOOP: DAD D
DCR C
JNZ MULTLOOP
MULTDONE: SHLD 4522h
HLT
```

Starting Address : 4300

Comments :

**User Data Grid**

| Address | Data |
|---|---|
| 4520 | FB |
| 4521 | 07 |
| 4522 | DD |
| 4523 | 06 |

**Hex Code Grid**

| Address | Data |
|---|---|
| 4300 | 3A |
| 4301 | 20 |
| 4302 | 45 |
| 4303 | 5F |
| 4304 | 16 |
| 4305 | 00 |
| 4306 | 3A |
| 4307 | 21 |
| 4308 | 45 |
| 4309 | 4F |
| 430A | 21 |
| 430B | 00 |
| 430C | 00 |
| 430D | 79 |
| 430E | B7 |
| 430F | CA |
| 4310 | 17 |
| 4311 | 43 |

1.Data Transfer Group :

| | | | |
|---|---|---|---|
| MOV | MVI | LDA | STA |
| LHLD | SHLD | LXI | XCHG |
| LDAX | STAX | IN | OUT |

2.Arithmatic Group :

| | | | | |
|---|---|---|---|---|
| ADD | ADI | ADC | ACI | DAA |
| SUB | SUI | SBB | SBI | DAD |
| INR | DCR | INX | DCX | |

3.Logical Group :

| | | | | |
|---|---|---|---|---|
| ANA | ANI | XRA | XRI | CMA |
| ORA | ORI | CMP | CPI | CMC |
| RLC | RRC | RAL | RAR | STC |

4.Branch and Machine Control Group

| | | | | |
|---|---|---|---|---|
| JMP | J_ | CALL | C_ | |
| RET | R_ | EI | DI | PUSH |
| PCHL | RST | RIM | SIM | POP |
| XTHL | SPHL | NOP | HLT | |

Auto Insert | Insert | Cancel

Instruction Information :

Code Pad